WoSM Academy

MAT231-F27 · CALCULUS II

Generating functions

Counting and solving recurrences

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Last time

- The Taylor series of f at a is $\sum \frac{f^{(k)}(a)}{k!}(x-a)^k$; it equals f when the remainder tends to 0.
- Series for $\frac{1}{1-x}$, e^x, sin, cos, $\ln(1+x)$, arctan, and the binomial series $(1+x)^\alpha = \sum \binom{\alpha}{k} x^k$.
- One function held a whole sequence: $\dfrac{1}{\sqrt{1-4x}} = \sum_{k=0}^{\infty} \binom{2k}{k} x^k$.

Today we turn that around: start from a sequence, and study it through one function.

This lecture

01 Generating functions

02 Operations on generating functions

03 Counting with generating functions

04 Solving linear recurrences

05 Catalan numbers

06 Asymptotics from generating functions (optional)

07 Summary and practice

Reading: H. S. Wilf, *generatingfunctionology*, Chapter 1.

01

Generating functions

A sequence in one function

> *"A generating function is a clothesline on which we hang up a sequence of numbers for display."*
>
> — *H. S. Wilf, generatingfunctionology*

- Encode a sequence $a_0, a_1, a_2, \ldots$ as one function $A(x) = \sum_{n \geq 0} a_n x^n$.
- Operations on sequences — shifting, summing, combining — become algebra on functions.
- This connects Discrete Structures (counting, recurrences, trees) with the power series of this module.

Ordinary generating functions

Definition

The **(ordinary) generating function** of a sequence $(a_n)_{n\geq 0}$ is $A(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$. We write $[x^n]\, A(x) = a_n$ for the coefficient of x^n.

Sequence	a_n	Generating function
constant	1	$\frac{1}{1-x}$
geometric	r^n	$\frac{1}{1-rx}$
binomial	$\binom{m}{n}$ (m fixed)	$(1+x)^m$
one more	$n+1$	$\frac{1}{(1-x)^2}$

A formula we will use often

Lemma

For every integer $k \geq 0$: $$\frac{1}{(1-x)^{k+1}} = \sum_{n=0}^{\infty} \binom{n+k}{k} x^n.$$

Proof: the binomial series (Lecture 6) with $\alpha = -(k+1)$ and x replaced by $-x$: $\binom{-(k+1)}{n}(-1)^n = \frac{(k+1)(k+2)\cdots(k+n)}{n!} = \binom{n+k}{k}$.

Example

$k = 0$: coefficients 1. $k = 1$: coefficients $n+1$. $k = 2$: coefficients $\binom{n+2}{2}$.

With $x \to rx$: $\frac{1}{(1-rx)^{k+1}} = \sum \binom{n+k}{k} r^n x^n$. After partial fractions, this reads off any rational generating function.

Formal power series

- We may treat $A(x)$ as a **formal power series**: a sequence written with a placeholder x, added and multiplied by the usual rules. No convergence is needed for the algebra.
- Convergence matters only when we *evaluate* A at a number, or use analysis: derivatives as limits, or the growth of a_n (Section 6).
- The series in this lecture have radius $R > 0$. Then A is a genuine function near 0, and by uniqueness (Lecture 5) it determines its coefficients.

> **Example**
>
> $\sum n!\, x^n$ has $R = 0$: useless as a function, but a perfectly good formal series.

02

Operations on generating functions

A dictionary

Let $A(x) \leftrightarrow (a_n)$ and $B(x) \leftrightarrow (b_n)$.

Operation	Sequence	Generating function
sum	$\alpha a_n + \beta b_n$	$\alpha A(x) + \beta B(x)$
shift right	$0, a_0, a_1, \dots$ (i.e. a_{n-1})	$xA(x)$
shift left	a_{n+1}	$\frac{A(x)-a_0}{x}$
scale	$c^n a_n$	$A(cx)$
multiply by n	$n\,a_n$	$xA'(x)$
partial sums	$a_0 + a_1 + \cdots + a_n$	$\frac{A(x)}{1-x}$
convolution	$\sum_{k=0}^{n} a_k b_{n-k}$	$A(x)\,B(x)$

Each line is a short check of coefficients. Partial sums: convolution with $\frac{1}{1-x}$.

Using the dictionary

- $n \leftrightarrow x \cdot \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{1-x} = \frac{x}{(1-x)^2}$, $\quad n^2 \leftrightarrow \frac{x(1+x)}{(1-x)^3}$ (Lecture 5)

Example (Sums of powers, without guessing)

- Partial sums of n: $\frac{x}{(1-x)^3} = x\sum_m \binom{m+2}{2}x^m$, so $\sum_{k=0}^{n} k = \binom{n+1}{2} = \frac{n(n+1)}{2}$.
- Partial sums of n^2: $\frac{x(1+x)}{(1-x)^4}$, so

$$\sum_{k=0}^{n} k^2 = \binom{n+2}{3} + \binom{n+1}{3} = \frac{n(n+1)(2n+1)}{6}.$$

In Discrete Structures you proved such formulas by induction, once you knew them. Here the formula comes out of the algebra.

Products count combined choices

If a_n counts the objects of one kind with size n, and b_n those of another kind, then

$$c_n = \sum_{k=0}^{n} a_k \, b_{n-k}$$

counts the *pairs* with total size n: choose sizes k and $n - k$, then one object of each. So $C(x) = A(x)\,B(x)$.

> **Example (Apples and bananas)**
>
> Bags of n fruits, using any number of apples and any number of bananas: $\frac{1}{1-x} \cdot \frac{1}{1-x} = \frac{1}{(1-x)^2}$, so there are $n + 1$ bags.

Identities for free

Example (Vandermonde's identity)

$(1+x)^m(1+x)^p = (1+x)^{m+p}$. Comparing coefficients of x^n:

$$\sum_{k=0}^{n} \binom{m}{k}\binom{p}{n-k} = \binom{m+p}{n}.$$

In words: choose n people from m women and p men, sorted by the number of women.

Example (Partial sums)

$\frac{(1+x)^m}{1-x}$ has coefficients $\sum_{k=0}^{n}\binom{m}{k}$; for $n \geq m$ this is $(1+1)^m = 2^m$.

03

Counting with generating functions

Making change

In how many ways can we pay n with coins worth 1, 2 and 5?

- Use the 2-coin $0, 1, 2, \ldots$ times: $1 + x^2 + x^4 + \cdots = \frac{1}{1-x^2}$. Likewise for the others.
- A payment is a choice for each coin, and the amounts add up:

$$\sum_{n \geq 0} w_n x^n = \frac{1}{1-x} \cdot \frac{1}{1-x^2} \cdot \frac{1}{1-x^5}.$$

n	0	1	2	3	4	5	6	7	8	9	10	11	12
w_n	1	1	2	2	3	4	5	6	7	8	10	11	13

Making change: the algorithm

How do we get the coefficients? Multiply in one factor at a time.

- If $B(x) = \frac{A(x)}{1-x^c}$, then $B(x) = A(x) + x^c B(x)$, that is

$$b_n = a_n + b_{n-c}.$$

```
ways = [1] + [0] * N          # the generating function 1
for c in (1, 2, 5):           # multiply by 1 / (1 - x^c)
    for n in range(c, N + 1):
        ways[n] += ways[n - c]
```

This is the standard dynamic-programming solution of the coin-change problem. The generating function explains *why* it is correct.

Solutions of $x_1 + x_2 + x_3 = n$

Each variable contributes a factor listing its allowed values.

- $x_i \geq 0$: $\frac{1}{(1-x)^3}$, so $\binom{n+2}{2}$ solutions.
- $x_i \geq 1$: $\frac{x^3}{(1-x)^3}$, so $\binom{n-1}{2}$ solutions.

Both agree with the stars-and-bars count of Discrete Structures.

Example ($x_1 \in \{0,1\}$, x_2 **even**, $x_3 \geq 0$)

$$(1+x) \cdot \frac{1}{1-x^2} \cdot \frac{1}{1-x} = \frac{1}{(1-x)^2} \quad \Longrightarrow \quad n+1 \text{ solutions.}$$

The constraints make a direct count awkward; the generating function handles them mechanically — and here it even simplifies.

Sequences of choices

In how many ways can you climb n stairs, taking 1 or 2 steps at a time?

- One move: $x + x^2$. A climb is a sequence of k moves, for some $k \geq 0$:

$$\sum_{k \geq 0} (x + x^2)^k = \frac{1}{1 - x - x^2} = 1 + x + 2x^2 + 3x^3 + 5x^4 + 8x^5 + \cdots$$

- The Fibonacci numbers appear. Why — and what is a formula for them? That is the next section.

04

Solving linear recurrences

The method

1. Write the recurrence, the values of n for which it holds, and the initial values.
2. Multiply by x^n and sum over those n.
3. Write each sum in terms of $A(x)$ (shifts!), and solve for $A(x)$.
4. Decompose into partial fractions (Calculus I, Unit 22).
5. Read off the coefficients:

$$\frac{1}{1-rx} \leftrightarrow r^n, \qquad \frac{1}{(1-rx)^{k+1}} \leftrightarrow \binom{n+k}{k} r^n.$$

Fibonacci numbers

$F_0 = 0,\ F_1 = 1,\ F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Let $F(x) = \sum F_n x^n$.

$$\sum_{n\geq 2} F_n x^n = \sum_{n\geq 2} F_{n-1} x^n + \sum_{n\geq 2} F_{n-2} x^n$$

$$F(x) - F_0 - F_1 x = x\big(F(x) - F_0\big) + x^2 F(x)$$

$$F(x) - x = xF(x) + x^2 F(x)$$

$$F(x) = \frac{x}{1 - x - x^2}.$$

The stairs of the last section: $\frac{1}{1-x-x^2} = \frac{F(x)}{x}$, so there are F_{n+1} ways to climb n stairs.

Binet's formula

Factor: $1 - x - x^2 = (1 - \varphi x)(1 - \psi x)$ with $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$ (check: $\varphi + \psi = 1$, $\varphi\psi = -1$).
Partial fractions, with $\varphi - \psi = \sqrt{5}$:

$$\frac{x}{(1 - \varphi x)(1 - \psi x)} = \frac{1}{\sqrt{5}}\left(\frac{1}{1 - \varphi x} - \frac{1}{1 - \psi x}\right).$$

Theorem (Binet's formula)

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}} \quad \text{for all } n \geq 0.$$

As $|\psi| < 1$, the second term is below $\frac{1}{2}$: F_n is $\frac{\varphi^n}{\sqrt{5}}$ rounded to the nearest integer.

The same answer from Linear Algebra

In Linear Algebra (Unit 18) you wrote the recurrence with a matrix:

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = M^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

and diagonalised M. Its eigenvalues are the roots of $\lambda^2 - \lambda - 1$: φ and ψ.

- The denominator $1 - x - x^2 = x^2\left(\frac{1}{x^2} - \frac{1}{x} - 1\right)$ has the roots $\frac{1}{\varphi}$ and $\frac{1}{\psi}$: the *reciprocals* of the eigenvalues.
- Two routes to one formula. Generating functions need no matrices, and handle extra terms in the recurrence easily (next).

A non-homogeneous recurrence: the Tower of Hanoi

Moving n discs: move $n-1$ to the spare peg, the largest to the target, then the $n-1$ on top. So $a_0 = 0$ and $a_n = 2a_{n-1} + 1$ for $n \geq 1$.

$$\sum_{n\geq 1} a_n x^n = 2x \sum_{n\geq 1} a_{n-1} x^{n-1} + \sum_{n\geq 1} x^n \quad \Longrightarrow \quad A(x) = 2xA(x) + \frac{x}{1-x}.$$

$$A(x) = \frac{x}{(1-x)(1-2x)} = \frac{1}{1-2x} - \frac{1}{1-x} \quad \Longrightarrow \quad a_n = 2^n - 1.$$

Check by induction (Discrete Structures, Unit 5): $2(2^{n-1} - 1) + 1 = 2^n - 1$.

Repeated roots

$a_0 = 1,\ a_1 = 6,\ a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$.

- As before: $A(x)\,(1 - 4x + 4x^2) = a_0 + (a_1 - 4a_0)\,x = 1 + 2x$, and $1 - 4x + 4x^2 = (1 - 2x)^2$.
- Partial fractions: $\dfrac{1 + 2x}{(1 - 2x)^2} = \dfrac{-1}{1 - 2x} + \dfrac{2}{(1 - 2x)^2}$.
- So $a_n = -2^n + 2(n + 1)2^n = (2n + 1)\,2^n$: $1, 6, 20, 56, 144, \ldots$

In general

For $a_n = c_1 a_{n-1} + \cdots + c_d a_{n-d}$, $A(x) = \frac{P(x)}{Q(x)}$ with $Q(x) = 1 - c_1 x - \cdots - c_d x^d$ and $\deg P < d$. If $Q(x) = \prod_i (1 - r_i x)^{m_i}$, then $a_n = \sum_i p_i(n)\, r_i^n$ with each p_i a polynomial of degree less than m_i. A double root gives terms $n\, r^n$.

05

Catalan numbers

Binary trees

A **binary tree** is empty, or a root with a left and a right subtree, both binary trees (Discrete Structures, Unit 15: trees). Let b_n be the number with n nodes.



The $b_3 = 5$ binary trees with 3 nodes.

For $n \geq 1$: a root, a left subtree with k nodes and a right subtree with $n - 1 - k$:

$$b_0 = 1, \qquad b_n = \sum_{k=0}^{n-1} b_k \, b_{n-1-k} \quad (n \geq 1). \qquad 1, 1, 2, 5, 14, 42, \ldots$$

A quadratic equation

The sum is a convolution: b_n is the coefficient of x^{n-1} in $C(x)^2$, where $C(x) = \sum b_n x^n$. Multiply by x^n and sum over $n \geq 1$:

$$C(x) - 1 = x\,C(x)^2.$$

Solve $xC^2 - C + 1 = 0$ for C:

$$C(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}.$$

Which sign? We need $C(0) = b_0 = 1$. With +, the numerator tends to 2 as $x \to 0$ and C blows up. With −, the limit is 1 (one step of L'Hôpital, or the series below). So

$$C(x) = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

Expanding $\sqrt{1-4x}$

Lecture 6: $\frac{1}{\sqrt{1-4x}} = \sum_{k\geq 0}\binom{2k}{k}x^k$ for $|x| < \frac{1}{4}$. As $\left(\sqrt{1-4x}\right)' = \frac{-2}{\sqrt{1-4x}}$, integrate term by term (Lecture 5):

$$\sqrt{1-4x} = 1 - 2\sum_{k\geq 0}\binom{2k}{k}\frac{x^{k+1}}{k+1}.$$

So $C(x) = \frac{1-\sqrt{1-4x}}{2x} = \sum_{n\geq 0}\frac{1}{n+1}\binom{2n}{n}x^n$.

Theorem (Catalan numbers)

The number of binary trees with n nodes is $C_n = \frac{1}{n+1}\binom{2n}{n}$: $1, 1, 2, 5, 14, 42, 132, 429, \ldots$

Catalan numbers everywhere

The same recurrence $C_n = \sum_k C_k C_{n-1-k}$ counts, for example:

- **balanced bracket sequences** with n pairs. Each one is $(A)B$ with A and B balanced: the first bracket and its partner split it, like a root and two subtrees.

$$n = 3: \quad ((())) \quad (()()) \quad (())() \quad ()(()) \quad ()()()$$

- ways to fully parenthesise a product $a_0 a_1 \cdots a_n$ of $n+1$ factors;
- triangulations of a convex polygon with $n+2$ vertices;
- lattice paths from $(0,0)$ to (n,n) with unit steps right and up that never go above the diagonal.

Same recurrence, same initial value, same numbers: one generating function solves them all.

06

Asymptotics from generating functions (optional)

The radius predicts the growth

- Informally: the growth of a_n is decided by the singularity of $A(x)$ *closest to* 0. Its distance is the radius R, and a_n grows like $\left(\frac{1}{R}\right)^n$, up to a slower factor.
- Precisely: $\limsup_n \sqrt[n]{|a_n|} = \frac{1}{R}$ (Cauchy–Hadamard, Lecture 5).

> **Example (Fibonacci)**
>
> $\frac{x}{1-x-x^2}$ is singular at $\frac{1}{\varphi} \approx 0.618$ and $\frac{1}{\psi} \approx -1.618$. The closer one gives $R = \frac{1}{\varphi}$, so F_n grows like φ^n — as Binet's formula says.

Catalan numbers grow like 4^n

$C(x) = \frac{1-\sqrt{1-4x}}{2x}$ is singular where $1 - 4x = 0$: $R = \frac{1}{4}$. So C_n grows like 4^n, up to a slower factor. In fact $C_n \sim \dfrac{4^n}{\sqrt{\pi}\, n^{3/2}}$:

n	C_n	$C_n \Big/ \frac{4^n}{\sqrt{\pi}\, n^{3/2}}$
10	16 796	0.898
20	6 564 120 420	0.946
50	$\approx 1.98 \cdot 10^{27}$	0.978
100	$\approx 8.97 \cdot 10^{56}$	0.989

The factor $n^{-3/2}$ comes from the *type* of singularity, a square root. This is the subject of analytic combinatorics (P. Flajolet, R. Sedgewick, *Analytic Combinatorics*, 2009). Stirling's formula, next lecture, proves the $\binom{2n}{n}$ part.

07

Summary and practice

Summary

- The generating function of (a_n) is $A(x) = \sum a_n x^n$; as a formal power series it needs no convergence.
- Shifts, scaling, $x\frac{\mathrm{d}}{\mathrm{d}x}$, partial sums $(\div(1-x))$ and convolution (product) translate operations on sequences.
- Products count combined choices: change-making, solutions of $x_1 + x_2 + x_3 = n$.
- Linear recurrences: multiply by x^n, sum, solve for A, partial fractions, read off coefficients. A double root gives $n\,r^n$.
- Catalan numbers from $C = 1 + xC^2$: $C_n = \frac{1}{n+1}\binom{2n}{n}$.
- The radius of A predicts the exponential growth of a_n.

Project 1 (series and generating functions) builds on this lecture.

Practice

1. Find the generating function: (a) $3^n + (-1)^n$ (b) $n\,2^n$ (c) $1, 0, 1, 0, 1, \ldots$ (d) $\binom{n+3}{3}$
2. Find a_n: (a) $\frac{1}{(1-3x)^2}$ (b) $\frac{x^2}{(1-x)^4}$ (c) $\frac{1+x}{1-x}$
3. In how many ways can you pay 20 with coins worth 1, 2, 5 and 10?
4. How many solutions does $x_1 + x_2 + x_3 = n$ have with $x_1 \geq 1$, $x_2 \geq 2$, $x_3 \geq 0$?
5. Solve $a_0 = 2$, $a_1 = 5$, $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$.
6. Solve $a_0 = 1$, $a_n = 3a_{n-1} + 2^n$ for $n \geq 1$.
7. Show that $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$.
8. Show that the balanced bracket sequences with n pairs satisfy the Catalan recurrence.
9. (Optional) Predict the growth of a_n if $A(x) = \frac{1}{1-2x-x^2}$.

Answers

1. (a) $\frac{1}{1-3x} + \frac{1}{1+x}$ (b) $\frac{2x}{(1-2x)^2}$ (c) $\frac{1}{1-x^2}$ (d) $\frac{1}{(1-x)^4}$
2. (a) $(n+1)\,3^n$ (b) $\binom{n+1}{3}$ (c) $a_0 = 1$, and $a_n = 2$ for $n \geq 1$
3. 40: the coefficient of x^{20} in $\frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})}$.
4. $\frac{x \cdot x^2}{(1-x)^3}$, so $\binom{n-1}{2}$.
5. $A(x) = \frac{2-5x}{(1-2x)(1-3x)} = \frac{1}{1-2x} + \frac{1}{1-3x}$, so $a_n = 2^n + 3^n$.
6. $A(x)(1-3x) = \frac{1}{1-2x}$, $A(x) = \frac{3}{1-3x} - \frac{2}{1-2x}$, so $a_n = 3^{n+1} - 2^{n+1}$.
7. Compare the coefficients of x^n in $(1+x)^n(1+x)^n = (1+x)^{2n}$, using $\binom{n}{k} = \binom{n}{n-k}$.
8. A non-empty sequence is $(A)B$, where the first bracket closes after A; if A has k pairs, B has $n-1-k$.
9. The root of $1 - 2x - x^2$ closest to 0 is $\sqrt{2} - 1$, so a_n grows like $(1+\sqrt{2})^n$.

Next time

Lecture 08 — Stirling's approximation (Unit 8)

- $n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$, from the sum $\ln n! = \sum_{k=1}^{n} \ln k$
- How good the approximation is
- Uses: $\log_2 n! \approx n \log_2 n$ in sorting, and $\binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}}$

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.